

**metanet\_param\_fit()**

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**Parameters.**

- $N$  : number of observed time samples,
- $S$  : number of calibrated segments,
- $T, \ell$  : time-step duration (h), segment length (km),
- $\lambda_s$  : fixed number of lanes in segment  $s$ ,
- $\hat{v}_{t,s}$  : observed velocity (km/h),
- $\hat{R}_{t,s}$  : observed total density across lanes (veh/km),
- $d_t$  : prescribed upstream flow across all lanes (veh/h),
- $R_t^{\text{ds}}$  : prescribed downstream total density (veh/km).

All  $0 \leq t < N$  and  $0 \leq s < S$ ; dynamics apply for  $1 \leq t < N$ . The argument `initial_traffic_state` supplies  $d_t$ .

**Model variables.**

| Variable               | Meaning                                 | Bounds               | Indices |
|------------------------|-----------------------------------------|----------------------|---------|
| $\eta_s^{\text{high}}$ | anticipation ( $\text{km}^2/\text{h}$ ) | $[10, 90]$           | $s$     |
| $\tau_s$               | relaxation time (h)                     | $[10/3600, 60/3600]$ | $s$     |
| $K_s$                  | density offset (veh/km/lane)            | $[5, 60]$            | $s$     |
| $\rho_s^{\text{crit}}$ | critical density (veh/km/lane)          | $[15, 95]$           | $s$     |
| $v_s^{\text{f}}$       | free-flow speed (km/h)                  | $[70, 150]$          | $s$     |
| $a_s$                  | fundamental-diagram exponent            | $[0.5, 5]$           | $s$     |
| $v_{t,s}$              | velocity (km/h)                         | $[10^{-3}, 150]$     | $t, s$  |
| $R_{t,s}$              | total density (veh/km)                  | $[10^{-3}, 400]$     | $t, s$  |

With `warmstart`, the bounds become  $\rho_s^{\text{crit}} \in [15, 100]$  and  $v_s^{\text{f}} \in [70, 140]$ . Lane counts  $\lambda_s$  are fixed, not fitted.

**Ramp variables.**  $\beta_s$  is the off-ramp split fraction;  $r_{t,s}$  is on-ramp inflow (veh/h), with  $r_{t,s} \equiv r_s$  for static ramps.

| Option                | Off-ramp bounds              | On-ramp bounds                |
|-----------------------|------------------------------|-------------------------------|
| Ramping, static       | $\beta_s \in [10^{-3}, 0.9]$ | $r_s \in [10^{-3}, 2000]$     |
| Ramping, time-varying | $\beta_s \in [10^{-3}, 0.6]$ | $r_{t,s} \in [10^{-3}, 2000]$ |
| No ramping            | $\beta_s = 0$                | $r_s = 0$                     |

If `ramp_mapping` is supplied with ramping enabled, initially fix

$$r_{t,s} = 10^{-3} \quad (s \notin \text{on-ramps}),$$

$$\beta_s = 10^{-3} \quad (s \notin \text{off-ramps}).$$

With time-varying ramps and a mapping, `fixed_inflows` then fixes selected  $r_{t,s} = r_{t,s}^{\text{given}}$ . Later warm-start or `x0` assignments can overwrite fixed values.

**Initial conditions.**

$$v_{0,s} = \hat{v}_{0,s}, \quad R_{0,s} = \hat{R}_{0,s}.$$

**Density dynamics.** Define the mainline inflow

$$q_{t-1,s}^{\text{in}} = \begin{cases} d_{t-1} & s = 0, \\ R_{t-1,s-1} v_{t-1,s-1} & s > 0. \end{cases}$$

$$R_{t,s} = R_{t-1,s} + \frac{T}{\ell} \left( q_{t-1,s}^{\text{in}} - \frac{R_{t-1,s} v_{t-1,s}}{1 - \beta_s} + r_{t,s} \right).$$

The ramp term uses  $t$ , not  $t - 1$ ; density  $R$  is total across lanes.

**Velocity dynamics.** Define per-lane density, downstream density, and upstream velocity:

$$\rho_{t,s} = \frac{R_{t,s}}{\lambda_s},$$

$$\rho_{t,s}^{\text{next}} = \begin{cases} R_{t,s+1}/\lambda_{s+1} & s < S - 1, \\ R_t^{\text{ds}}/\lambda_s & s = S - 1, \end{cases}$$

$$v_{t,s}^{\text{prev}} = \begin{cases} v_{t,s} & s = 0, \\ v_{t,s-1} & s > 0. \end{cases}$$

With fundamental-diagram speed

$$V_s(\rho) = v_s^f \exp \left[ -\frac{1}{a_s} \left( \frac{\rho}{\rho_s^{\text{crit}}} \right)^{a_s} \right].$$

impose

$$v_{t,s} = v_{t-1,s} + \frac{T}{\tau_s} (V_s(\rho_{t-1,s}) - v_{t-1,s})$$

$$+ \frac{T}{\ell} v_{t-1,s} (v_{t-1,s}^{\text{prev}} - v_{t-1,s})$$

$$- \frac{\eta_s^{\text{high}} T}{\tau_s \ell} \frac{\rho_{t-1,s}^{\text{next}} - \rho_{t-1,s}}{\rho_{t-1,s} + K_s}.$$

There is no VSL minimum or smooth velocity floor in this fitter.

**Optional A regularizer.** When `use_A_regularizer=True`, define the implemented entries

$$\begin{aligned}
V'_s(\rho) &= -\frac{V_s(\rho)}{\rho_s^{\text{crit}}} \left( \frac{\rho}{\rho_s^{\text{crit}}} \right)^{a_s-1}, \\
b_{t,s}^{3,\text{self}} &= \frac{T}{\tau_s} V'_s(\rho_{t,s}) + \frac{\eta_s^{\text{high}} T}{\tau_s \ell} \frac{\rho_{t,s}^{\text{next}} + K_s}{(\rho_{t,s} + K_s)^2}, \\
b_{t,s}^{3,\text{down}} &= -\mathbf{1}_{s < S-1} \frac{\eta_s^{\text{high}} T}{\tau_s \ell (\rho_{t,s} + K_s)}, \\
b_{t,s}^{4,\text{self}} &= 1 - \frac{T}{\tau_s} + \frac{T}{\ell} (v_{t,s}^{\text{prev}} - 2v_{t,s}), \\
b_{t,s}^{4,\text{up}} &= \mathbf{1}_{s > 0} \frac{T}{\ell} v_{t,s}.
\end{aligned}$$

Then

$$\mathcal{R}_A = \sum_{t=1}^{N-1} \sum_{s=0}^{S-1} \left[ (b_{t,s}^{3,\text{self}})^2 + (b_{t,s}^{3,\text{down}})^2 + (b_{t,s}^{4,\text{self}})^2 + (b_{t,s}^{4,\text{up}})^2 \right].$$

These are the code's regularizer entries, not the exact Jacobian in  $(R, v)$ .

**Previous-parameter penalty.** If `prev_param_path` is supplied, denote its values by superscript `prev` and define

$$\begin{aligned}
\theta_s &= (\eta_s^{\text{high}}, \tau_s, K_s, \rho_s^{\text{crit}}, v_s^{\text{f}}, a_s, \beta_s, r_s), \\
c &= (90, 60/3600, 60, 100, 150, 5, 0.9, 2000), \\
\mathcal{R}_{\text{prev}} &= 10 \sum_{s=0}^{S-1} \sum_{j=1}^8 \left( \frac{\theta_{s,j} - \theta_{s,j}^{\text{prev}}}{c_j} \right)^2.
\end{aligned}$$

This penalty currently requires static ramps (or ramping disabled).

**Objective.** Let  $\hat{v}_{\max} = \max_{t,s} \hat{v}_{t,s}$ ,  $\hat{R}_{\max} = \max_{t,s} \hat{R}_{t,s}$ , and  $\lambda_{\text{reg}} = \text{lambda\_reg}$ . Minimize

$$\min \sum_{t=0}^{N-1} \sum_{s=0}^{S-1} \left[ 20 \left( \frac{v_{t,s} - \hat{v}_{t,s}}{\hat{v}_{\max}} \right)^2 + \left( \frac{R_{t,s} - \hat{R}_{t,s}}{\hat{R}_{\max}} \right)^2 \right] + \mathcal{R}_{\text{prev}} + \lambda_{\text{reg}} \mathcal{R}_A.$$

Omit each penalty when its corresponding option is disabled.